

1. Definition

Amazon Virtual Private Cloud (VPC) is a virtual network in AWS that allows users to launch AWS resources in a logically isolated environment.

2. Key Features

- **Complete network isolation from other AWS users**
- **Customizable IP address range (CIDR block)**
- **Control over subnets, route tables, gateways**
- **Secure access using security groups and network ACLs**

3. Components of VPC

- **Subnets**
- **Route Tables**
- **Internet Gateway**
- **NAT Gateway**
- **Security Groups**

4. Working

- **User creates a VPC with an IP range**
- **Resources like EC2 are launched inside VPC**
- **Traffic is controlled using routing and security rules**

5. Advantages

- **High security and isolation**
- **Flexible network configuration**
- **Scalable infrastructure**

6. Example

A company creates a VPC to host its web application, keeping database servers private and web servers public.

7. Conclusion

VPC provides secure and customizable cloud networking for AWS resources.

1. Definition : A subnet is a subdivision of a VPC's IP range where resources are placed.

2. Types of Subnets

- **Public Subnet:** Has access to the internet
- **Private Subnet:** No direct internet access

3. Characteristics

- Each subnet belongs to one Availability Zone
- Uses a portion of VPC's IP range
- Controlled by route tables

4. Working

- VPC is divided into multiple subnets
- Public subnet connects to Internet Gateway
- Private subnet connects via NAT (optional)

5. Diagram

VPC (10.0.0.0/16)

|

|----- **Public Subnet (10.0.1.0/24) → Web Server**

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|----- **Private Subnet (10.0.2.0/24) → Database Server**

6. Advantages

- **Better network organization**
- **Improved security separation**
- **Efficient resource management**

7. Example

Web servers are placed in public subnet, while sensitive databases are placed in private subnet.

8. Conclusion

Subnets help in segmentation and security control within VPC.

1. Definition

A route table is a set of rules that determines how network traffic is directed within a VPC.

2. Components

- **Destination (IP range)**
- **Target (gateway, instance, etc.)**

3. Working

- **When data is sent, route table checks destination**
- **Matches route and forwards traffic to target**

4. Types

- **Main Route Table**
- **Custom Route Table**

5. Example Entry

Destination	Target
0.0.0.0/0	Internet Gateway
10.0.0.0/16	Local

6. Example

If a user accesses a website, route table directs traffic to Internet Gateway.

7. Conclusion : Route tables control traffic flow and connectivity in VPC.

Internet Gateway

1. Definition : An Internet Gateway (IGW) is a component that allows communication between a VPC and the internet.

2. Functions

- **Enables incoming and outgoing internet traffic**
- **Performs NAT (Network Address Translation) for instances**

3. Working

- **IGW is attached to VPC**
- **Route table must have route to IGW**
- **Public subnet instances can access internet**

4. Requirements

- **Public IP or Elastic IP required**
- **Proper route table configuration**

5. Diagram

EC2 Instance → Route Table → Internet Gateway → Internet

6. Advantages

- **Enables external communication**
- **Easy to configure**
- **Highly available (managed by AWS)**

7. Example

A web server in public subnet uses IGW to serve web pages to users worldwide.

1. Definition

Amazon S3 is a scalable object storage service used to store and retrieve any amount of data over the internet.

Amazon S3 is a cloud-based object storage service provided by AWS.

It allows users to store and retrieve any amount of data from anywhere using the internet.

Data is stored in the form of objects inside buckets.

It provides high durability, scalability, and security for data storage.

2. Key Features

- **Unlimited storage capacity**
- **High durability (99.999999999%)**
- **High availability**
- **Pay-as-you-go pricing**
- **Supports multiple storage classes**

3. Working

- **Data stored as objects inside buckets**
- **Each object has key, metadata, and content**
- **Access via URL using HTTP/HTTPS**

4. Storage Classes

- **S3 Standard (frequent access)**
- **S3 Intelligent-Tiering**
- **S3 Glacier (archival)**

5. Advantages

- **Backup and disaster recovery**
- **Big data analytics storage**
- **Media hosting**

6. Example

Companies like Netflix store media files for streaming.

1. Definition

- **Object Storage:** Stores data as objects (data + metadata + key)
- **Bucket:** Logical container to store objects
- **S3 uses an object storage model where data is stored as objects.**
Each object contains data, metadata, and a unique key.
A bucket is a container used to store these objects.
Buckets help in organizing and managing data efficiently.

2. Components of Object

- **Data (file)**
- **Metadata (info about file)**
- **Key (unique identifier)**

3. Bucket Features

- **Globally unique name**
- **Region specific**
- **Can store unlimited objects**
- **Supports versioning and policies**

4. Working

- **Create bucket → Upload objects → Access via URL**

5. Diagram

Bucket (project-data)

|
|--- image.png
|--- report.pdf
|--- video.mp4

6. Advantages

- **Easy organization of files**
- **Fast retrieval**
- **Highly scalable**

7. Example

Storing student assignments in a university system.

1. Definition : Security ensures authorized access and data protection in S3 buckets. Encryption methods are used to protect stored data. It helps in maintaining data confidentiality and integrity.

2. Security Methods

- **IAM Policies (user-level access)**
- **Bucket Policies (resource-level control)**
- **ACLs (basic permissions)**

3. Encryption

- **Server-Side Encryption (SSE-S3, SSE-KMS)**
- **Client-Side Encryption**

4. Access Control

- **Block public access feature**
- **Use of roles and permissions**

5. Additional Features

- **MFA (Multi-Factor Authentication) for delete**
- **Logging for monitoring access**

6. Example

Only admin can delete files, users can only view files.

Web Hosting

1. Definition

S3 can be used to host static websites.

It allows uploading of HTML, CSS, and JavaScript files.

Websites can be accessed using a public URL.

It is a cost-effective and serverless hosting solution.

2. Features

- **Supports HTML, CSS, JavaScript**
- **Provides public URL**
- **No need for server management**

3. Steps

- 1. Create bucket**
- 2. Upload website files**
- 3. Enable static website hosting**
- 4. Set public permissions**

4. Advantages

- **Low cost**
- **Easy deployment**
- **Highly scalable**

5. Example

Hosting a company portfolio or landing page.

1. Definition

Logging records all access requests made to S3 buckets. Event notifications trigger actions when changes occur in objects. These events can be integrated with AWS services like Lambda or SNS. It helps in monitoring and automation of tasks.

- **Logging:** Records access details
- **Events:** Trigger actions on changes

2. Logging

- Tracks request details (who accessed, when)
- Used for auditing and security

3. Event Notifications

- Triggered on:
 - Object upload
 - Object delete

4. Integration

- AWS Lambda (automatic processing)
- SNS (notifications)
- SQS (queue system)

5. Advantages

- Real-time processing
- Automation of workflows
- Monitoring user activity

6. Example : When a file is uploaded, Lambda resizes the image automatically.

Cross Region Replication

1. Definition

Cross Region Replication (CRR) copies data from one S3 bucket to another in a different region. Cross Region Replication copies data between S3 buckets in different regions. It works automatically and requires versioning to be enabled. It improves data availability and disaster recovery. It ensures backup and global access to data.

2. Features

- Automatic replication
- Asynchronous process
- Requires versioning enabled

3. Working --> Source bucket → Destination bucket

--> Data copied across regions

4. Advantages

- Disaster recovery
- High availability
- Reduced latency for global users

5. Diagram ---→ S3 Bucket (India) —→ S3 Bucket (USA)

6. Example : Backup data from India region to US region.

1. Glacier

Amazon S3 Glacier

Definition:

Glacier is a low-cost storage service used for long-term data archiving.

Features:

- **Very low storage cost**
- **Suitable for rarely accessed data**
- **Retrieval time: minutes to hours**

Use Case / Example:

Storing old company records or backup files that are not frequently used.

2. Versioning

Definition:

Versioning is a feature that keeps multiple versions of an object in an S3 bucket.

Features:

- **Protects against accidental deletion**
- **Allows recovery of previous versions**
- **Automatically stores updated copies**

Working:

- **When a file is updated → new version is created**
- **Old version is still available**

Example:

If a user edits a document, previous versions can be restored anytime.

3. Lifecycle Policy

Definition:

Lifecycle policy is used to automatically manage objects by moving or deleting them after a certain time.

Features:

- **Moves data to cheaper storage (e.g., Glacier)**
- **Deletes old or unused data**
- **Fully automated**

Working Example:

- **After 30 days → move to Glacier**
 - **After 1 year → delete object**
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4. Advantages

- **Reduces storage cost**
- **Improves data management**
- **Provides backup and recovery**
- **Automates data lifecycle**